Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



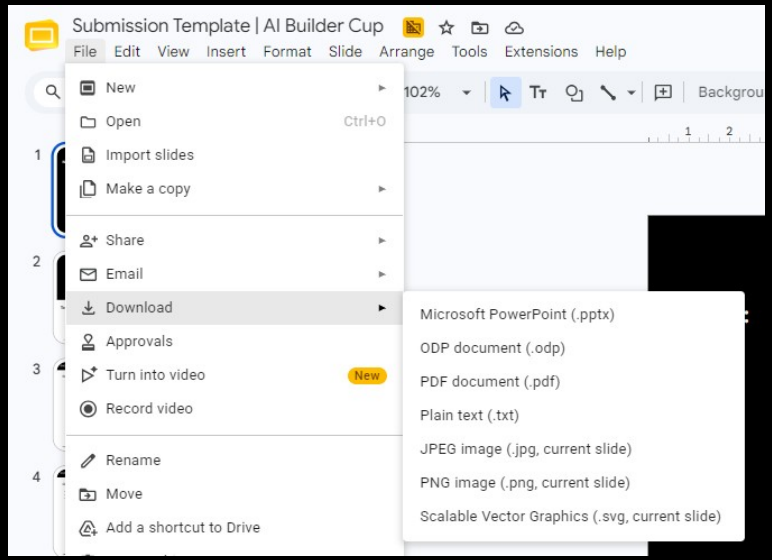
Team Details

Team name: Vaultkeepers

Team leader name: Santosh Ramesh

Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

Solution: Financial Friday



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

Brief about the idea

Financial Friday

A financial buddy that turns statement history into a plan you can follow.

People can see past transactions, yet still struggle to decide what they can afford next. Friday connects reviewed history to a conversation about their next goal.



One supported statement. One conversation. A visual cash roadmap with clear assumptions.

India-first financial assistance. No bank credentials or payment authority.

Opportunities

How different is it from existing ideas?

Friday joins reviewed statement explanations and conversational goal planning in one private workspace. Exact code checks each visual cash plan before the user saves it.

How will it solve the problem?

A supported import organises transaction history. The user confirms budget facts in chat, then Friday tests the goal and its alternatives against that same budget.

USP of the proposed solution

Gemini handles language and documents. An independent INR calculation engine preserves the reserve, debt commitments and every monthly rupee. Users can inspect the assumptions.

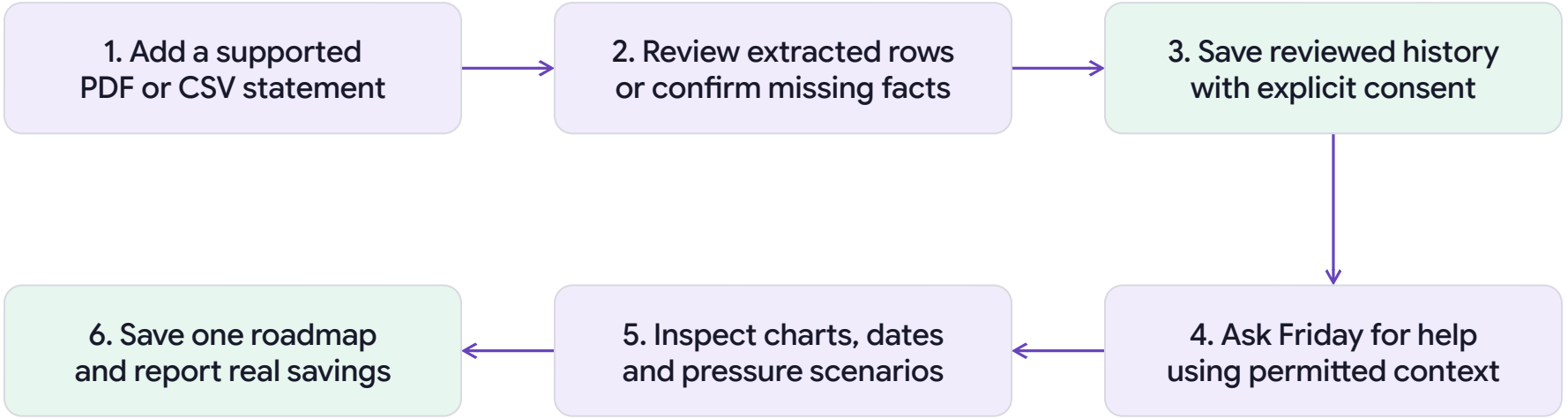
Combination and execution are the differentiation. No world-first or patent claim.

List of features offered by the solution

- Supported PDF/CSV statements with preview, source checks and reviewed categories.
- Spending charts, transaction references and candidates for the user to review.
- Chat-led purchase, savings and emergency-reserve goals with missing-fact questions.
- Exact allocations, dated checkpoints, pressure scenarios and feasible alternatives.
- Private saved roadmap, self-reported savings updates and Undo.
- Optional Google voice turns and attributed public research, with explicit controls.

Current consumer journey: statements, money overview, goals and Ask Friday.

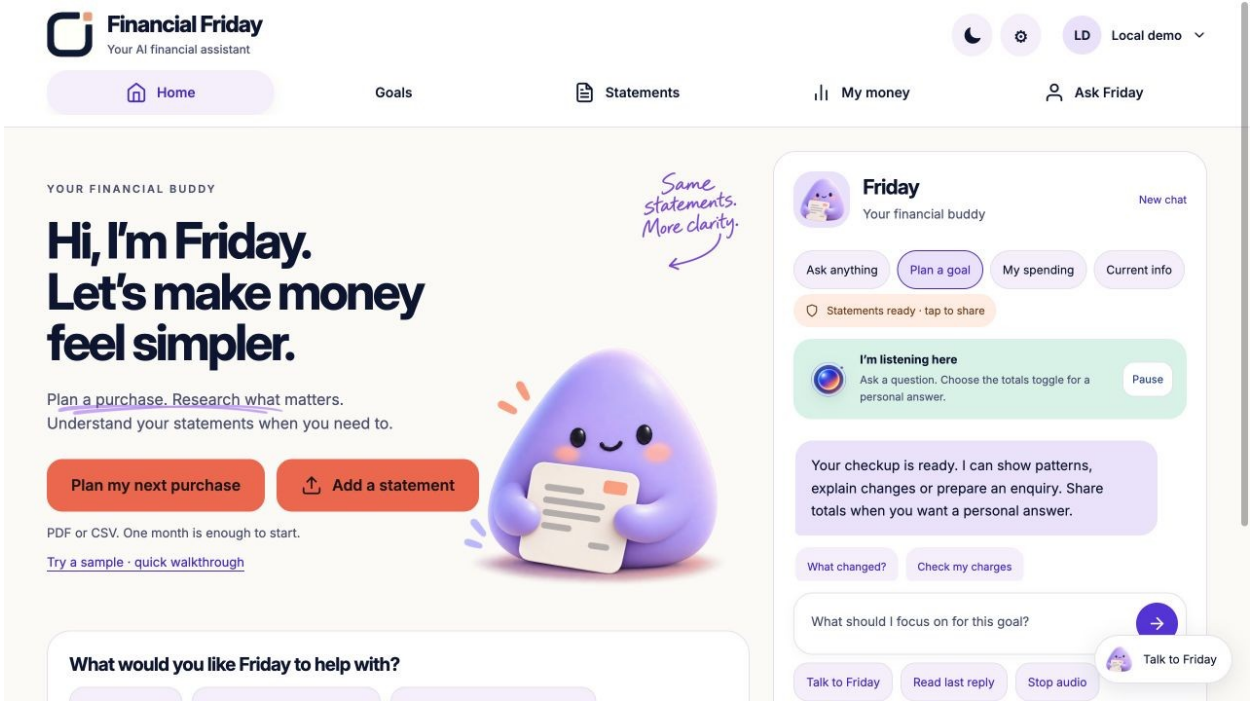
Process flow diagram or Use-case diagram



AI interprets the request. Exact code checks the result. A failed source check or missing fact returns to review.

A user can plan a goal without importing a statement. No money moves.

Wireframes/Mock diagrams of the proposed solution (optional)



Implemented UI

Clear starting actions

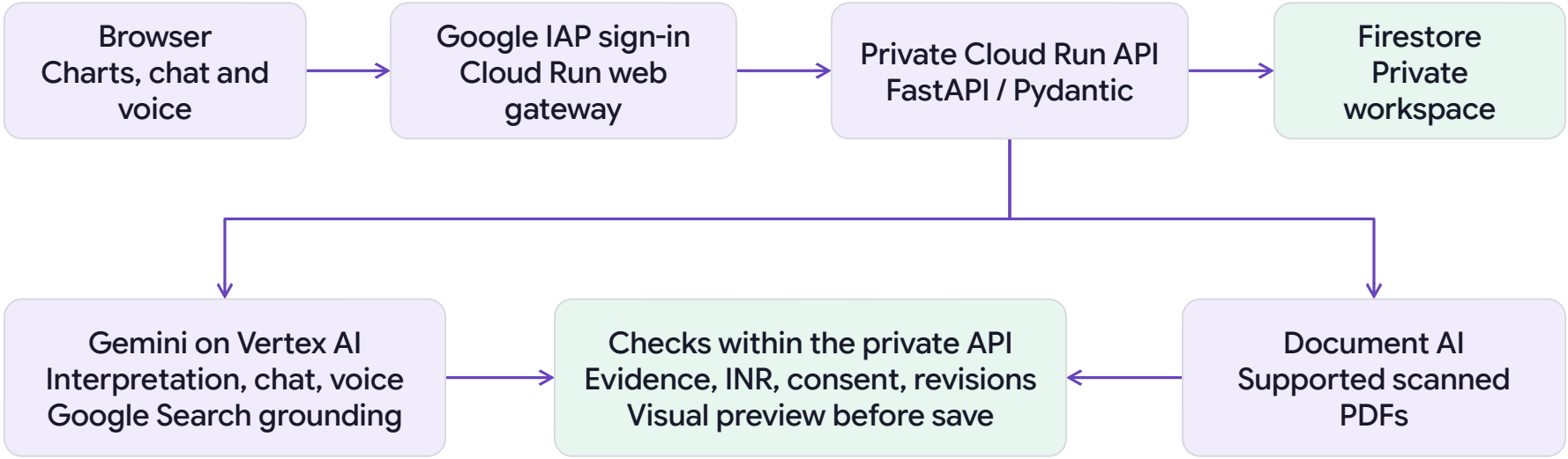
Persistent Friday chat

Visible voice controls

Account and privacy menu

Actual local product screen. Artificial workspace only.

Architecture diagram of the proposed solution



Reviewed history and goals persist per user. Private totals reach Gemini only when the user permits them.

Public research excludes private statement data. Raw uploads are not archived.

Technologies to be used in the solution

Technology	Implemented purpose
Cloud Run	Web gateway and private Python API, deployed in asia-south1.
Gemini on Vertex AI	PDF interpretation, structured goal intake, grounded chat and voice.
Firestore	Tenant-scoped reviewed rows, preferences, roadmap state and revisions.
Document AI	OCR path for supported scanned statement PDFs.
IAM / IAP / Secret Manager	Google identity, service boundaries and server-side configuration.
Cloud Build / Artifact Registry	Container builds and immutable deployment images.
HTML / CSS / JS / Python	Responsive UI, FastAPI, Pydantic, local Docker and SQLite.

Gemini models power the AI. Firebase Hosting/Auth, BigQuery and model fine-tuning are not used.

Estimated implementation cost (optional)

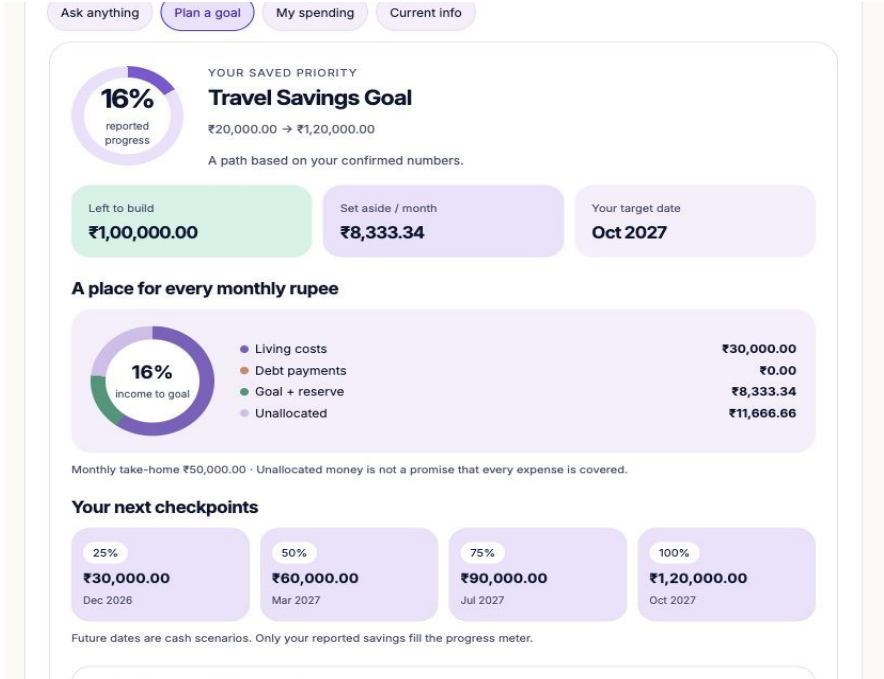
Usage-based prototype on a Google Cloud trial

Cost driver	Current control
PDF, chat, research and voice calls	Explicit opt-in, request limits and bounded context.
Known-format CSV analysis	Deterministic parsing and calculations need zero model calls.
Cloud Run and Firestore operations	Serverless services and bounded workspace sizes.
Repeated speech playback	Page-memory replay avoids another model call when cached.

A production per-user estimate still needs measured traffic, tokens, document pages and retention. Trial credits do not make the service permanently free.

No fabricated spend, savings or cost-per-user claim.

Snapshots of the prototype



A plan the user can inspect

Artificial travel goal	INR
Target in 12 months	1,20,000
Starting pot after reserve	20,000
Monthly saving step	8,333.34
Protected cash reserve	50,000

Friday also shows pressure cases and budget alternatives. Progress changes only when the user reports additional savings.

Actual product output. Scenario values are not forecasts or verified bank balances.

Prototype Performance report/Benchmarking

Evaluation scope	Recorded result
Native regression suite	868 run: 867 passed, 1 optional integration skipped.
Container regression suite	868 run: 818 passed, 50 environment-dependent skips.
Offline cloud tooling checks	68 passed.
Isolated real-Google acceptance	8 calls across voice, Search and goal persistence flows.
Authored 39-case review benchmark	66.67% precision, 73.68% recall, 38.89% case-level FPR.

Software tests establish tested behaviour, not financial efficacy. The synthetic review benchmark still has misses and false warnings. Independent real-bank validation remains open.

Sources: CURRENT_RELEASE.json, FINAL_POLISH_0390.md and VALIDATION_HANDOFF.md in the public repository.

Additional Details/Future Development (if any)

Next validation priorities

- Independent, authorised bank-layout and explanation evaluation.
- Measure task completion, incorrect guidance and warning fatigue.
- Test multilingual use, accessibility and account-policy variations.
- Assess consented bank integrations only through authorised partners.

Current boundaries

- Supported INR PDF/CSV limits apply. Unknown evidence stays unknown.
- No bank feeds, live balances, payment execution or automatic money recovery.
- Education and cash planning exclude personalised securities trades.
- Savings progress is self-reported. No guaranteed returns or outcomes.

Goal: a measured consumer pilot before claims of broad banking readiness.

Provide links to your:

GitHub Public Repository

https://github.com/Santxsh27/financial-friday

Final Product Link

https://financial-friday-web-171681243260.asia-south1.run.app/#home

Demo Video Link (3–4 minutes)

Video link will be supplied separately in the submission form after recording.

Walkthrough and artificial statement downloads

https://financial-friday-web-171681243260.asia-south1.run.app/assets/friday-judge-guide.html

Product requires Google sign-in. Source and judge samples are public.

Judge quick start

One sample. A visible result. Your own private workspace.

- Open the product and sign in with a Google account. Check My account.
- Use Try a sample, then add one six-month artificial CSV and review the rows.
- Compare 42 entries, INR3,00,000 in and INR1,64,694 out before saving.
- Ask Friday about spending, or describe a goal and continue with Gemini.
- Review the visual roadmap. Confirm saving only if you want to keep it.

The judge kit contains five header families for the same invented activity. Test one at a time. Bank labels describe illustrative layouts, not certified exports.

Never upload real private banking records for a public demonstration.

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Thank you

